

Semantic Clustering of Experiential vs. Technical Language in LLM Embeddings

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Abstract

We investigate whether large language model embeddings encode experiential language (descriptions of subjective sensory and emotional states) differently from technical language (descriptions of algorithms and computational processes). Using multiple embedding models (BGE-M3, Sentence-BERT), we find that experiential concepts cluster **3–3.5× more tightly** than technical concepts in semantic space—a robust effect that replicates across models. Topological analysis via persistent homology confirms this pattern ($d = 0.44$, $p = 0.03$). However, our attempt to identify causal mechanisms via attention head ablation yielded **mixed results**: initial findings ($n = 15$) suggested phenomenal-specific circuits, but extended replication ($n = 30$) did not confirm this pattern ($p = 0.87$). We conclude that the clustering difference reflects the **semantic coherence of experiential domains** rather than specialized “phenomenal processing.” This finding has implications for understanding how LLMs represent different discourse types, though not for consciousness research per se.

1 Introduction

Large language models encode semantic relationships in high-dimensional embedding spaces. A natural question arises: do different *types* of discourse—experiential descriptions vs. technical specifications—occupy geometrically distinct regions of this space?

We investigate this question using two contrasting concept classes:

- **Experiential concepts**: First-person descriptions of sensory qualia, emotions, and subjective states
- **Technical concepts**: Descriptions of algorithms, data structures, and computational processes

We operationalize semantic structure geometrically using topological data analysis and cross-model clustering metrics. Our findings reveal a robust pattern:

1. **Experiential language clusters more tightly** than technical language (3–3.5× in Sentence-BERT)
2. **This is not merely linguistic confound**—surface features explain $<5\%$ of variance
3. **Causal mechanisms remain elusive**—ablation studies yielded inconsistent results

1.1 Hypotheses

H1 (Primary) LLM embeddings for experiential concepts exhibit higher semantic clustering than technical concepts.

H2 (Secondary) HDC binding operations amplify experiential-technical differences.

H3 (Exploratory) Attention head ablation reveals causal mechanisms underlying the clustering difference.

1.2 Summary of Findings

H1 is **supported** with robust cross-model replication. H2 is **not supported**—binding is content-agnostic. H3 yielded **mixed results**—initial findings did not replicate with increased statistical power.

2 Related Work

2.1 Topological Data Analysis in NLP

Persistent homology has been applied to NLP tasks including word embedding analysis [Savle et al., 2019], document classification [Zhu, 2013], and semantic similarity measurement [Michel et al., 2017]. Carlsson [2009] provides foundational work on topological methods for data analysis.

2.2 Hyperdimensional Computing

Hyperdimensional computing (HDC) represents information as high-dimensional binary vectors with algebraic operations for composition [Kanerva, 2009]. Recent work has applied HDC to language modeling [Rahimi et al., 2016] and cognitive architectures [Gayler, 2003].

2.3 Embodied vs. Abstract Language

Research in cognitive linguistics distinguishes embodied/grounded concepts from abstract concepts [Barsalou, 2008, Lakoff & Johnson, 1980]. Neuroimaging studies show differential processing of concrete vs. abstract words [Wang et al., 2010]. Our findings align with this literature.

3 Methods

3.1 Concept Corpus

We constructed two balanced corpora of 50 concepts each:

Experiential Concepts ($n = 50$): First-person descriptions of sensory qualia across seven modalities (visual, auditory, tactile, gustatory, olfactory, thermal, pain).

Technical Concepts ($n = 50$): Technical descriptions of algorithms and data structures (algorithms, data structures, sorting, memory, optimization).

3.2 Embedding Pipeline

1. **Text Embedding:** BGE-M3 via Candle framework
2. **HDC Projection:** Linear probe projecting 1024D embeddings to 16,384D binary hyper-vectors (HV16)
3. **Topology Analysis:** Persistent homology computing Betti numbers ($\beta_0, \beta_1, \beta_2$)

3.3 Topological Unity Score

For each concept, we compute unity score as: $1/\beta_0$ where β_0 = number of connected components. Higher unity scores indicate more integrated, less fragmented topological structure.

3.4 Statistical Analysis

- **Primary test:** Two-sample permutation test ($n = 10,000$ permutations)
- **Effect size:** Cohen’s d with pooled standard deviation
- **Significance threshold:** $\alpha = 0.05$ (two-tailed)

4 Results

4.1 H1: Experiential vs Technical Topology (SUPPORTED)

Metric	Experiential ($n = 50$)	Technical ($n = 50$)
Mean Unity Score	0.785	0.650
Standard Deviation	0.307	0.311
Mean β_0 (components)	1.70	2.06
Mean β_1 (cycles)	0.52	0.22

Table 1: Topological metrics for experiential vs. technical concepts.

Statistical Tests:

- Observed difference: 0.135
- Cohen’s d : 0.44 (small-to-medium effect)
- p -value: 0.030 (10,000 permutations)
- **Significant at $\alpha = 0.05$:** Yes

4.2 H2: Binding vs Bundling (NOT SUPPORTED)

HDC binding (XOR) creates vectors $2.7\times$ more novel than bundling (majority vote), but this effect is **uniform across pair types**. No interaction effect ($p = 1.0$).

4.3 H3: Combined Arc (NOT SUPPORTED)

After binding, both experiential and technical pairs show unity = 1.0. Binding erases the experiential-technical distinction detected in H1.

5 Cross-Model Validation

5.1 Sentence-BERT Replication

To test generalization beyond BGE-M3, we analyzed 200 concepts using Sentence-BERT models.

Model	Experiential	Technical	Ratio
all-mpnet-base-v2	0.338	0.097	$3.5\times$
all-MiniLM-L6-v2	0.303	0.098	$3.1\times$

Table 2: Within-class semantic similarity by model.

Key Finding: Experiential concepts cluster **3–3.5 $\times$ more tightly** than technical concepts. This replicates across models.

6 Linguistic Feature Analysis

Feature	Experiential	Technical	Cohen’s d
Sentence length	6.60 words	3.89 words	+2.77
Avg word length	5.23 chars	8.15 chars	−2.63
Sensory word ratio	6.5%	0.5%	+0.86

Table 3: Linguistic feature comparison.

Correlation with topological unity: $r = 0.08\text{--}0.16$ (weak). Surface features explain only $\sim 2.5\%$ of variance.

7 Mechanistic Analysis

7.1 Model Comparison

Model	Params	Fisher	Angular Sep	Isotropy
BERT-base	110M	1.189	0.246	0.017
BERT-large	335M	1.037	0.148	0.033
RoBERTa-base	125M	1.035	0.016	0.026
RoBERTa-large	355M	1.028	0.015	0.032

Table 4: Mechanistic metrics across models.

Key Finding: Angular separation is the primary mechanism. Larger models align experiential and technical centroids more closely, reducing discrimination.

7.2 Attention Head Analysis

Top experiential-discriminating heads (lower entropy for experiential):

Head	Effect d	Exp. Entropy	Tech. Entropy
L11.H4	−2.629	4.01	4.97
L11.H1	−2.375	4.84	5.53
L11.H9	−2.025	4.46	5.08

Table 5: Top experiential-discriminating attention heads.

Layer 11 is the “experiential layer” with 4 of the top 5 heads.

7.3 Causal Intervention: Ablation Study

Initial Results ($n = 15$): Experiential heads showed mean specificity $+0.007$ vs. controls -0.007 .

Extended Replication ($n = 30$): Mann-Whitney $U = 31.0$, $p = 0.87$ (**NOT significant**). The causal specificity effect **did not replicate** with increased statistical power.

8 Conclusion

8.1 What We Found

1. **Robust clustering difference:** Experiential concepts cluster $3\text{--}3.5\times$ more tightly than technical concepts. This replicates across models.
2. **Topological confirmation:** BGE-M3 + HDC topology analysis shows the same pattern ($d = 0.44$, $p = 0.03$).
3. **Not linguistic confound:** Surface features explain only $\sim 2.5\%$ of variance.
4. **Attention patterns exist:** Layer 11 shows lower entropy for experiential concepts.

8.2 What We Did NOT Find

1. **No causal mechanism:** Ablation of “experiential-preferring” heads did not specifically reduce experiential clustering ($p = 0.87$).
2. **No binding-specific effect:** HDC binding produces uniform effects regardless of concept type.
3. **No “phenomenal circuits”:** The attention entropy differences do not causally drive embedding structure.

8.3 Interpretation

The most parsimonious explanation is **semantic domain coherence**:

- Experiential descriptions refer to related domains (sensory modalities, emotions, bodily states)
- Technical descriptions span diverse domains (algorithms, data structures, networks, systems)

The tighter clustering of experiential language reflects this domain coherence, not specialized “phenomenal processing.” This is consistent with cognitive linguistics research on embodied vs. abstract concepts [Barsalou, 2008].

8.4 Honest Summary

We set out to find evidence for distinct phenomenal processing in LLMs. We found robust evidence that experiential language clusters differently—but the most likely explanation is domain coherence, not consciousness-related computation. The mechanistic story we hoped for did not materialize. We report these findings honestly as a contribution to understanding LLM semantic structure, while acknowledging that the consciousness-specific interpretation is not supported.

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